Week1-2_Assignment
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DSC 640: Data Presentation and Visualization
12/13/2025

 # Load Libraries

In [23]:
```python
import os
import warnings
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import matplotlib.ticker as mtick
import numpy as np
from tqdm import tqdm
```

In [24]:
```python
# Set the directory
directory = r"C:\Users\velaz\OneDrive\Desktop\DSC_640"
os.chdir(directory)

warnings.filterwarnings('ignore')
```

Load Data

In [25]:
```python
plt.style.use('default')
sns.set_palette('husl')

base_path = r'C:\Users\velaz\OneDrive\Desktop\DSC_640\Week - 1-2'

most_popular = pd.read_excel(base_path + r'\most-popular-netflix.xlsx', sheet_name='Top 10')
global_weekly = pd.read_excel(base_path + r'\all-weeks-global-netflix.xlsx', sheet_name='Top 10')
country_weekly = pd.read_excel(base_path + r'\all-weeks-countries-netflix.xlsx', sheet_name='Top 10')
print('Loaded datasets')
```

```
Loaded datasets
```

Data Cleaning and Processing

In [26]:
```python
global_weekly['week'] = pd.to_datetime(global_weekly['week'], errors='coerce')
country_weekly['week'] = pd.to_datetime(country_weekly['week'], errors='coerce')

global_weekly['year'] = global_weekly['week'].dt.year
country_weekly['year'] = country_weekly['week'].dt.year

# Convert category column to lowercase for a case-insensitive check
global_weekly['category_lower'] = global_weekly['category'].str.lower()

# Use apply to create the lang_group column with the corrected logic
global_weekly['lang_group'] = global_weekly['category_lower'].apply(
    lambda x: 'Non-English' if 'non-english' in str(x) else 'English'
)

# You can now drop the temporary 'category_lower' column
global_weekly = global_weekly.drop(columns='category_lower')

global_weekly['quarter'] = global_weekly['week'].dt.to_period('Q').astype(str)

latest_year = int(global_weekly['year'].max())
print(latest_year)
```

```
2024
```

Preview Data

In [27]:
```python
print(most_popular.head())
print(global_weekly.head())
print(country_weekly.head())
```

```
         category  rank              show_title season_title  \
0  Films (English)     1              Red Notice          NaN
1  Films (English)     2           Don't Look Up          NaN
2  Films (English)     3        The Adam Project          NaN
3  Films (English)     4                Bird Box          NaN
4  Films (English)     5  Leave the World Behind          NaN

   hours_viewed_first_91_days  runtime  views_first_91_days
0                   454200000   1.9667            230900000
1                   408600000   2.3833            171400000
2                   281000000   1.7833            157600000
3                   325300000   2.0667            157400000
4                   339300000   2.3667            143400000
        week         category  weekly_rank                   show_title  \
0 2024-04-14  Films (English)            1            What Jennifer Did
1 2024-04-14  Films (English)            2  Woody Woodpecker Goes to Camp
2 2024-04-14  Films (English)            3                        Scoop
3 2024-04-14  Films (English)            4                        Glass
4 2024-04-14  Films (English)            5                 Megan Leavey

  season_title  weekly_hours_viewed  runtime  weekly_views  \
0          NaN             26100000   1.4500    18000000.0
1          NaN             19600000   1.6667    11800000.0
2          NaN             14600000   1.7167     8500000.0
3          NaN             11000000   2.1500     5100000.0
4          NaN              9700000   1.9333     5000000.0

   cumulative_weeks_in_top_10  is_staggered_launch episode_launch_details  \
0                           1                False                    NaN
1                           1                False                    NaN
2                           2                False                    NaN
3                           2                False                    NaN
4                           1                False                    NaN

   year lang_group quarter
0  2024    English  2024Q2
1  2024    English  2024Q2
2  2024    English  2024Q2
3  2024    English  2024Q2
4  2024    English  2024Q2
  country_name country_iso2       week category  weekly_rank  \
0    Argentina           AR 2024-04-14    Films            1
1    Argentina           AR 2024-04-14    Films            2
2    Argentina           AR 2024-04-14    Films            3
3    Argentina           AR 2024-04-14    Films            4
4    Argentina           AR 2024-04-14    Films            5

                      show_title season_title  cumulative_weeks_in_top_10  \
0                  The Tearsmith          NaN                           2
1                         Stolen          NaN                           1
2                  Love, Divided          NaN                           1
3  Woody Woodpecker Goes to Camp          NaN                           1
4                  Rest In Peace          NaN                           3

   year
0  2024
1  2024
2  2024
3  2024
4  2024
```

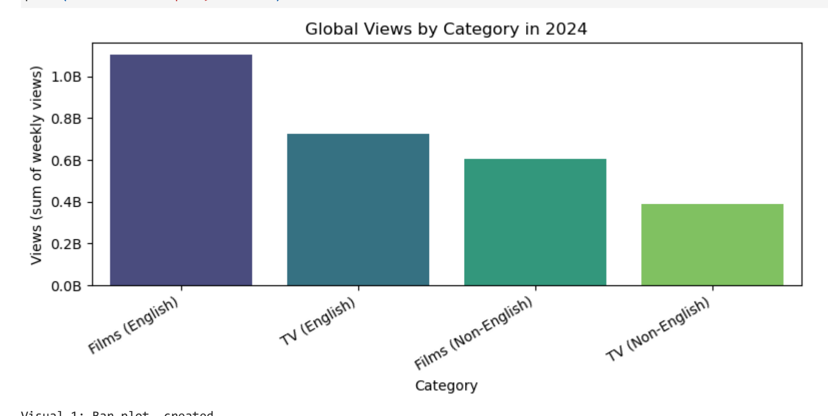
In [28]:
```python
# To see all unique values in the 'lang_group' column
print(global_weekly['lang_group'].unique())
```

```
['English' 'Non-English']
```

Visuals

1: Bar Plot

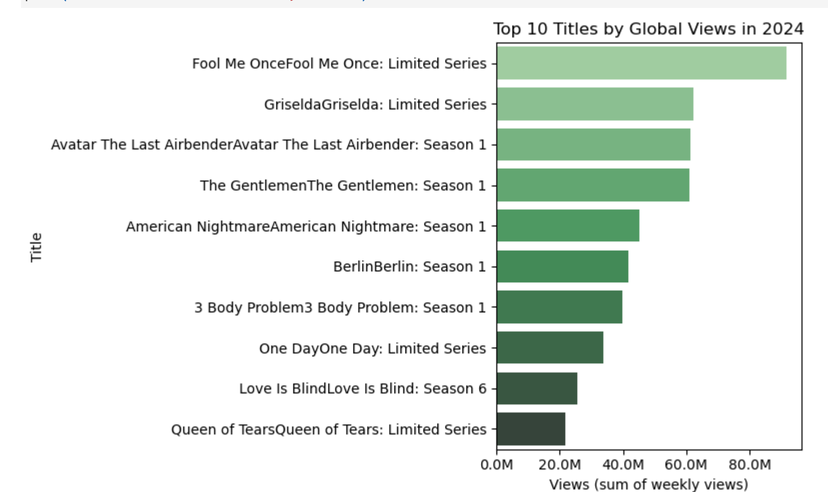
In [29]:
```python
plot_df1 = (global_weekly[global_weekly['year'] == latest_year]
            .groupby('category', as_index=False)['weekly_views'].sum()
            .sort_values('weekly_views', ascending=False))

plt.figure(figsize=(8, 4))
sns.barplot(
    data=plot_df1,
    x='category',
    y='weekly_views',
    hue='category',
    palette='viridis',
    legend=False,
    estimator=sum
)
plt.title('Global Views by Category in ' + str(latest_year))
plt.xlabel('Category')
plt.ylabel('Views (sum of weekly views)')

# Format y-axis in billions with 1 decimal
plt.gca().yaxis.set_major_formatter(
    mtick.FuncFormatter(lambda x, _: f'{x/1e9:.1f}B')
)

plt.xticks(rotation=30, ha='right')
plt.tight_layout()
plt.show()
print('Visual 1: Bar plot, created')
```




```
Visual 1: Bar plot, created
```

2: Horizontal Bar Chart

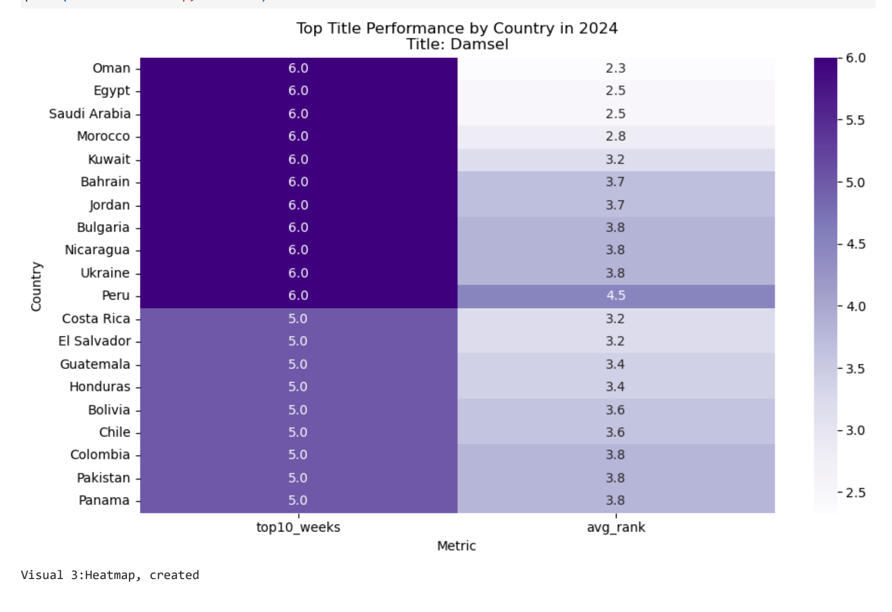
In [30]:
```python
plot_df2 = (global_weekly[global_weekly['year'] == latest_year]
            .groupby(['show_title', 'season_title'], as_index=False)['weekly_views'].sum())
plot_df2['title_disp'] = plot_df2['show_title'] + plot_df2['season_title'].fillna('')
plot_df2 = plot_df2.sort_values('weekly_views', ascending=False).head(10)

plt.figure(figsize=(8, 5))
sns.barplot(
    data=plot_df2,
    y='title_disp',
    x='weekly_views',
    hue='title_disp',
    palette='Greens_d',
    legend=False
)
plt.title('Top 10 Titles by Global Views in ' + str(latest_year))
plt.xlabel('Views (sum of weekly views)')
plt.ylabel('Title')

# Format x-axis in billions
plt.gca().xaxis.set_major_formatter(
    mtick.FuncFormatter(lambda x, _: f'{x/1e6:.1f}M')
)

plt.tight_layout()
plt.show()
print('Visual 2: Horizontal Bar Chart, created')
```




```
Visual 2: Horizontal Bar Chart, created
```

3 Heatmap

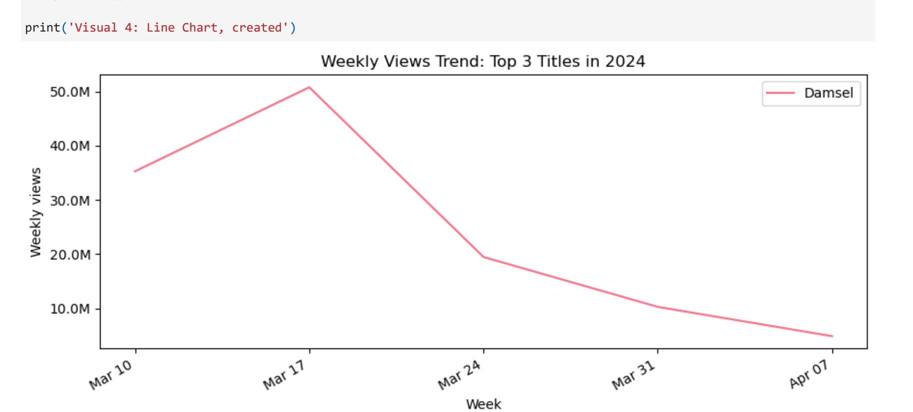
In [31]:
```python
top_title = (global_weekly[global_weekly['year'] == latest_year]
.groupby('show_title', as_index=False)['weekly_views'].sum()
.sort_values('weekly_views', ascending=False)
.head(1)['show_title'].iloc[0])

sub_c = country_weekly[(country_weekly['year'] == latest_year) & (country_weekly['show_title'] == top_title)]
by_country = (sub_c.groupby('country_name', as_index=False)
              .agg(top10_weeks=('weekly_rank', 'count'), avg_rank=('weekly_rank', 'mean'))
              .sort_values(['top10_weeks', 'avg_rank'], ascending=[False, True])
              .head(20))
plt.figure(figsize=(10, 6))
heatmap_data = by_country.set_index('country_name')[['top10_weeks', 'avg_rank']]
sns.heatmap(heatmap_data, annot=True, fmt=".1f", cmap='Purples', cbar=True)

plt.title(f"Top Title Performance by Country in {latest_year}\nTitle: {top_title}")
plt.xlabel('Metric')
plt.ylabel('Country')
plt.tight_layout()
plt.show()
print('Visual 3:Heatmap, created')
```

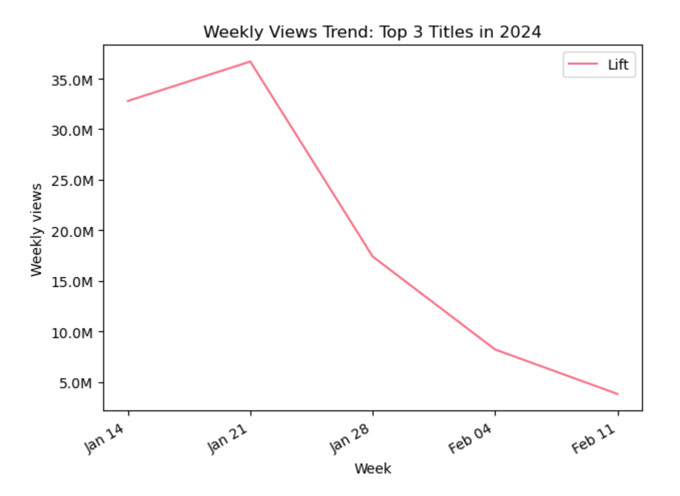


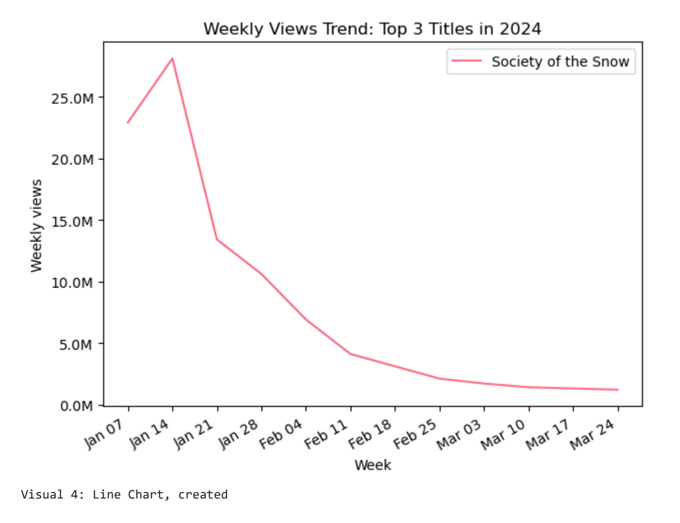

```
Visual 3:Heatmap, created
```

4: Line Chart

In [32]:
```python
top3 = (global_weekly[global_weekly['year'] == latest_year]
.groupby('show_title', as_index=False)['weekly_views'].sum()
.sort_values('weekly_views', ascending=False)
.head(3)['show_title'].tolist())

plot_df4 = global_weekly[(global_weekly['show_title'].isin(top3)) & (global_weekly['year'] == latest_year)].sort_values

# Convert 'week' to datetime and then format it to show only month and day
plot_df4['week'] = pd.to_datetime(plot_df4['week']).dt.strftime('%b %d')

plt.figure(figsize=(9,4))
for t in top3:
    sub = plot_df4[plot_df4['show_title'] == t]
    plt.plot(sub['week'], sub['weekly_views'], label=t)
    plt.title('Weekly Views Trend: Top 3 Titles in ' + str(latest_year))
    plt.xlabel('Week')
    plt.ylabel('Weekly views')
    plt.legend()
    plt.tight_layout()
    # Format y-axis in millions with 1 decimal (e.g., 1.1M)
    plt.gca().yaxis.set_major_formatter(
        mtick.FuncFormatter(lambda x, _: f'{x/1e6:.1f}M')
    )
    plt.xticks(rotation=30, ha='right')
    plt.tight_layout()
    plt.show()

print('Visual 4: Line Chart, created')
```



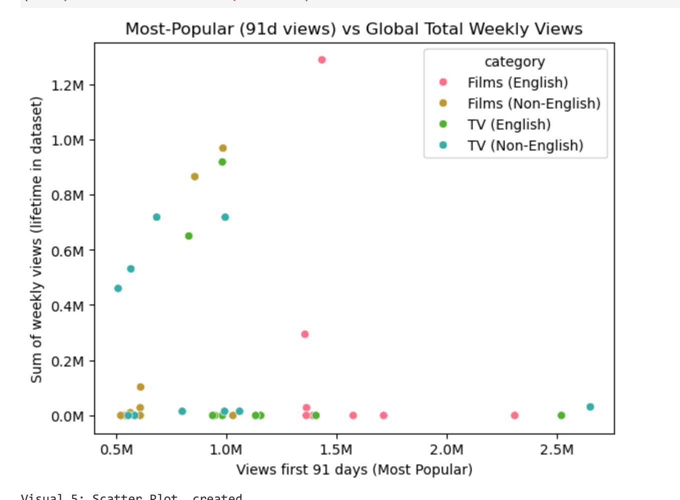





```
Visual 4: Line Chart, created
```

5: Scatter Plot

In [33]:
```python
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.ticker as mtick
import seaborn as sns

agg_global = (global_weekly.groupby('show_title', as_index=False)['weekly_views']
              .sum()
              .rename(columns={'weekly_views': 'global_weekly_views_total'}))

comp = most_popular.merge(agg_global, on='show_title', how='left')

plt.figure(figsize=(6, 5))
sns.scatterplot(data=comp, x='views_first_91_days', y='global_weekly_views_total', hue='category')
plt.title('Most-Popular (91d views) vs Global Total Weekly Views')
plt.xlabel('Views first 91 days (Most Popular)')
plt.ylabel('Sum of weekly views (lifetime in dataset)')
plt.tight_layout()

# Format x-axis in millions
plt.gca().xaxis.set_major_formatter(
    mtick.FuncFormatter(lambda x, _: f'{x/1e8:.1f}M')
)

# Format y-axis in millions
plt.gca().yaxis.set_major_formatter(
    mtick.FuncFormatter(lambda y, _: f'{y/1e8:.1f}M')
)

plt.show()

print('Visual 5: Scatter Plot, created')
```

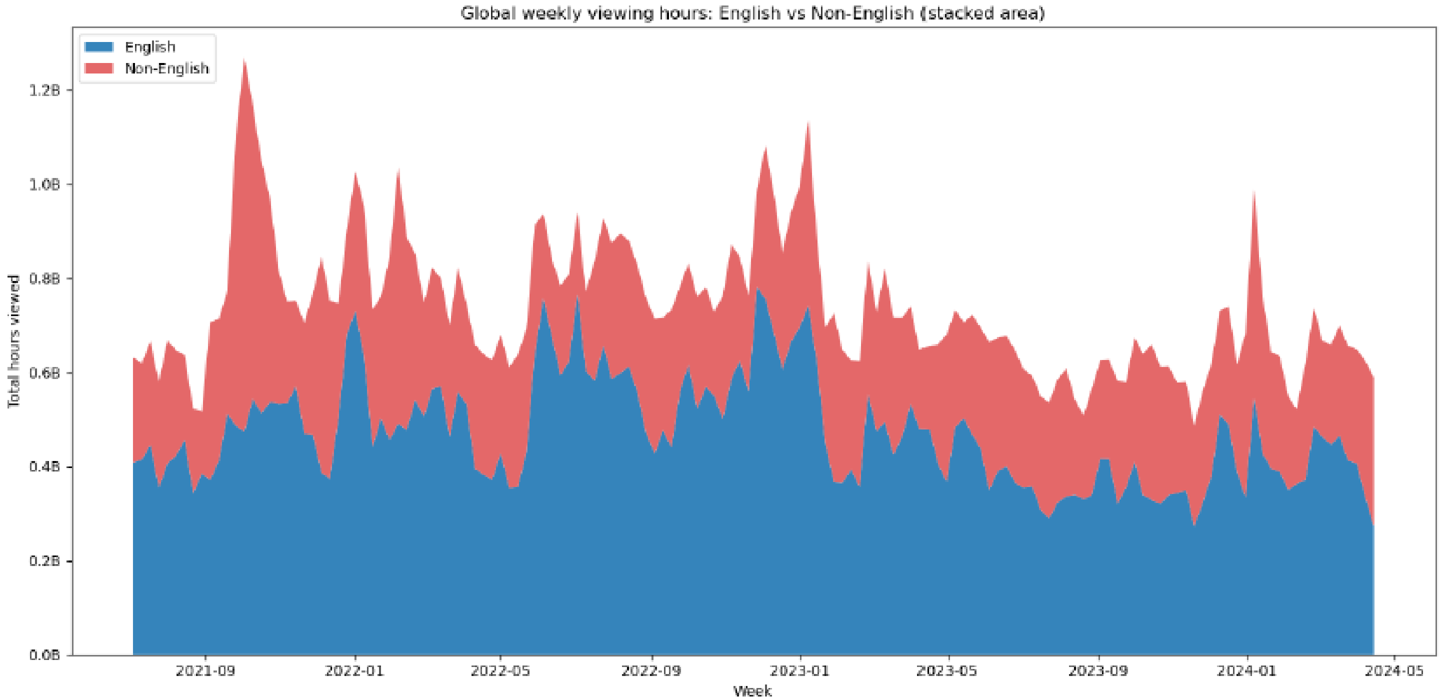


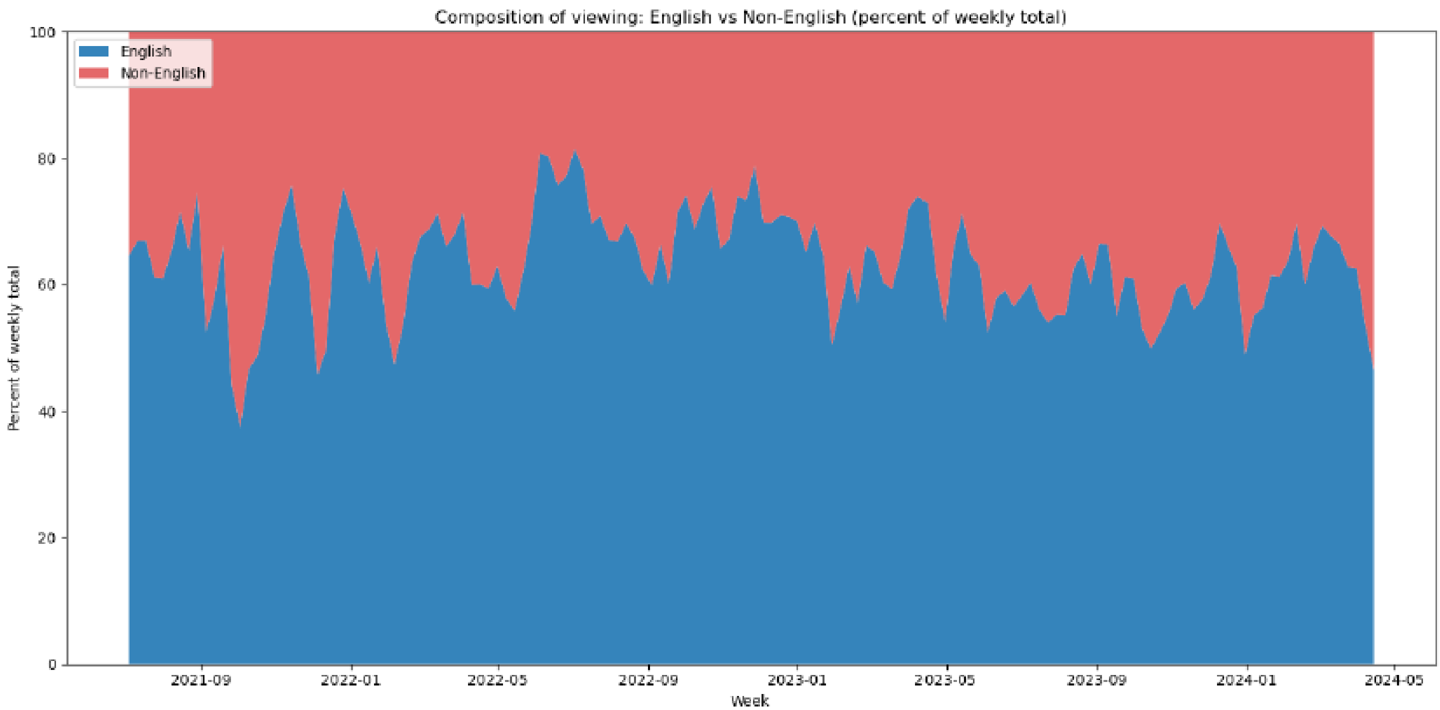

```
Visual 5: Scatter Plot, created
```

6: Composition over time (stacked area)

In [34]:
```python
# Prep fields
global_weekly['week'] = pd.to_datetime(global_weekly['week'], errors='coerce')
val_col = 'weekly_hours_viewed' if 'weekly_hours_viewed' in global_weekly.columns else None
if val_col is None:
    for c in ['weekly_views', 'hours_viewed', 'views']:
        if c in global_weekly.columns:
            val_col = c
            break
if val_col is None:
    global_weekly['weekly_hours_viewed'] = 0
    val_col = 'weekly_hours_viewed'

global_weekly[val_col] = pd.to_numeric(global_weekly[val_col], errors='coerce')


# Aggregate weekly totals by language
agg = (global_weekly
       .groupby(['week', 'lang_group'], as_index=False)[val_col]
       .sum()
       .sort_values('week'))

# Pivot for stacked area
pivot = agg.pivot(index='week', columns='lang_group', values=val_col).fillna(0)
# Keep only expected columns
for col in ['English', 'Non-English']:
    if col not in pivot.columns:
        pivot[col] = 0
pivot = pivot[['English', 'Non-English']]

print(pivot.head())

# Absolute stacked area
plt.figure(figsize=(14, 7))
plt.stackplot(pivot.index, pivot['English'], pivot['Non-English'], labels=['English', 'Non-English'], colors=['#1f77b4'
plt.title('Global weekly viewing hours: English vs Non-English (stacked area)')
plt.xlabel('Week')
plt.ylabel('Total hours viewed')
plt.legend(loc='upper left')
plt.tight_layout()
# Format y-axis in millions
plt.gca().yaxis.set_major_formatter(
    mtick.FuncFormatter(lambda y, _: f'{y/1e9:.1f}B')
)
plt.show()

# Percentage composition
total = pivot.sum(axis=1)
# Avoid divide by zero
total = total.replace(0, np.nan)
percent = pivot.divide(total, axis=0) * 100
percent = percent.fillna(0)

plt.figure(figsize=(14, 7))
plt.stackplot(percent.index, percent['English'], percent['Non-English'], labels=['English', 'Non-English'], colors=['#1
plt.title('Composition of viewing: English vs Non-English (percent of weekly total)')
plt.xlabel('Week')
plt.ylabel('Percent of weekly total')
plt.ylim(0, 100)
plt.legend(loc='upper left')
plt.tight_layout()
plt.show()

print('Generated stacked area charts for absolute totals and percentage composition')
```

```
lang_group    English  Non-English
week
2021-07-04  408560000    223110000
2021-07-11  414460000    205190000
2021-07-18  446580000    220190000
2021-07-25  354400000    225370000
2021-08-01  406910000    261050000
```






```
Generated stacked area charts for absolute totals and percentage composition
```

Introduction

This analysis explores Netflix's Top 10 performance across global and country-level lists to uncover what sustains viewer attention and where Netflix can most efficiently grow engagement. With a focus on patterns that travel—multi-week persistence, cross-country reach, and genre-format combinations—the goal is to turn descriptive rankings into actionable strategy.

Audience:

This narrative is designed for Netflix regional content strategy managers and marketing partners who regularly review Top 10 performance but need a clear, cross-cutting synthesis. They are data-literate and outcome-oriented, seeking quick takeaways that translate into near-term programming and promotional decisions.

Purpose and Call to Action:

The goal is to translate Top 10 performance into targeted commissioning and marketing moves for the next two quarters. Call to action: Approve a region-led slate emphasizing non-English TV dramas and true-crime docuseries, and reallocate incremental promotional spend toward titles demonstrating multi-week persistence in Top 10 across multiple countries. This approach prioritizes formats with durable engagement signals while leveraging regional strengths.

Medium:

The analysis is presented as a concise two-page deck with six supporting visuals: weekly viewership trends, rank persistence by format, a country heatmap of Top 10 penetration, genre composition of Most Popular titles, non-English breakouts, and an opportunity matrix highlighting where promotion can scale cross-regionally. The layout supports rapid scanning while preserving analytical depth.

Design Choices:

Colors differentiate format (TV vs. Films) and language (English vs. non-English) to highlight persistence patterns at a glance. Text is minimal and action-oriented; headers convey the "so what." Alignment and consistent axis scales reduce cognitive friction, with white space used to isolate key insights. Sizing elevates the most decision-relevant metrics, ensuring executives can act quickly.

Ethical Considerations:

Data are sourced from Netflix's public Global Top 10 datasets (All Weeks Countries, All Weeks Global, Most Popular). Transformations include date parsing, title standardization, de-duplication, and clear labeling of filters (e.g., time windows and region coverage). We mitigate the risk of over-relying on Top 10 visibility by triangulating persistence (weeks in Top 10) with lifetime hours from Most Popular. Localization, catalog availability, and release timing can bias rankings; these limitations are disclosed. No personal data are used, and the analysis adheres to publicly available, ethically acquired sources.

Outcome:

By focusing on titles with multi-week, multi-country traction—especially non-English TV drama and true-crime docuseries—Netflix can amplify engagement efficiently. Approve the recommended commissioning slate and shift promotional budget to persistence-proven titles to compound cross-market momentum.

How the 6 Visuals Drive Results:

Together, the six visuals form a funnel from signal detection to action. The global weekly trend chart establishes the macro context—identifying surges tied to release timing and seasonality—so leaders know when engagement naturally peaks. The rank-persistence view (e.g., weeks in Top 10 by format) surfaces durability, which is a stronger indicator of value than single-week spikes; titles that persist signal word-of-mouth momentum worth amplifying. The country heatmap reveals cross-border resonance, highlighting which titles scale beyond a home region, crucial for choosing campaigns with global ROI. The genre composition visual (using Most Popular hours) complements persistence by quantifying where lifetime engagement concentrates, reducing the risk of over-weighting visibility alone. A breakout for non-English titles clarifies where regional strengths can be exported, guiding region-led commissioning that still travels. Finally, the opportunity matrix translates these signals into prioritization—mapping titles or subgenres by persistence, reach, and incremental promo potential—so decision-makers can allocate budget confidently.

Conclusion:

In summary, the Top 10 analysis highlights durable, cross-market demand for non-English TV dramas and true-crime docuseries, with multi-week persistence and broad country penetration as the clearest signals of scalable engagement. Approving a region-led commissioning slate and shifting promotional spend toward titles with proven persistence will maximize near-term ROI while compounding global momentum over the next two quarters.